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Path planning and optimal control of a 4WS vehicle using calculus of variations

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In this paper, optimal control of a 4 wheel steering vehicle (4WSV) is performed using the calculus of variations, and its related optimal path is extracted according to a predefined objective function. Nowadays, 4WS vehicles are growing rapidly as a result of their excellent maneuverability. Considering the fact that these systems are over-constrained, possible trajectories through which the vehicle can be transferred between two definite boundaries are not unique. As a result, extracting the optimum path, by which a predefined cost function could be minimized, is a useful study. To meet this goal, firstly, the model of the system is represented. Afterward, the calculus of variations method is applied to control the vehicle in an optimal way. Implementing the optimal values of control inputs not only satisfies the boundary conditions but also minimizes a specific cost function, simultaneously. The cost function is set to minimize the error and energy consumption, and the corresponding optimal path is also calculated. In order to verify the correctness of modeling, the results of MATLAB simulation are compared with those of CarSim. The efficiency of the considered optimal control method is also investigated by conducting some analytic and comparative simulation scenarios in MATLAB. The superiority of the proposed controlling method compared to non-optimal strategies is demonstrated by comparing the corresponding cost functions. Finally, it is proven that modeling and implementing the proposed optimal control method results in the optimal path between two predetermined boundaries, by which the maximum accuracy and the minimum energy consumption can be acquired.

Four wheel steering vehicle (4WSV), Optimal control, Path planning, Calculus of variations method, Modeling

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1. Introduction

4 wheel steering vehicle (4WSV) is a new type of vehicle in which both the front and the rear wheels are independently controllable. Therefore, their maneuverability is potentially better than traditional vehicles. Such an increase in the number of control inputs provides the possibility of vehicle movement through various paths between two specific points. Thus, it is possible to extract the path which can satisfy certain constraints or minimize a required cost function. Although 2WS vehicles are controllable, they have some limitations in their movements, which require more time and energy compared to 4WS vehicles. This is contributed to the fact that since the number of actuating inputs can be increased employing 4WSV thus the system will be redundant and consequently it is possible to move the vehicle between two boundary positions using more trajectories amongst which some of them need a lower amount of torque. A typical example of such a phenomenon is parallel parking. To perform this operation, a more complicated path between two predetermined points needs to be traveled by the 2WS system compared to the 4WS one.

The optimal control method can be applied to a system adopting two different approaches, including closed-loop and open-loop strategies. Although closed-loop approaches have better robustness against the parametric uncertainties and external disturbances, the only parameter that can be

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optimized by them is the dynamics of the error. Whereas the main dynamics of the system also need to be optimized. Therefore, it can be concluded that closed-loop optimization methods are generally less efficient than open-loop strategies. In an open-loop control approach, the optimum path is derived using the calculus of variations, and the corresponding control input is derived by solving a set of simultaneous partial differential equations using the Hamiltonian function (Pontryagin). In this approach, there is no need to use the feedback signal of the states. Thus, the necessity of sensor implementation is canceled. Applying this optimal input to the system, the optimal states of the system can be calculated, through which the system moves from a predefined primary condition to a secondary one while a specific cost function is minimized.

In the research history of 4WS vehicles, investigating the kinematic and kinetic models of such systems formed the initially conducted studies. Since a 4WS vehicle can be considered firstly a mobile robot, investigating research about mobile robots can also be useful. Zhao and Bement [1] studied the controllability and stability of a non-holonomic robot, and the complete kinematic and dynamic models were presented using forward and inverse approaches. Spentzas et al. [2,3] studied 4WS and 2WS vehicle systems specifically and compared different forms of steerability in these two forms of vehicles. Also, a novel form of 4WS vehicles has been proposed using the bicycle model.

Dębowski et al. [4] showed the validity of the four-wheel steering (4WS) bicycle model and tested the response of the model to a lane change maneuver, using transfer functions and a "bang-bang" control signal. Additionally, Singh et al. [5] studied the differences between 4WS and 2WS vehicles and investigated the advantages and disadvantages of each type. They also studied the importance of using 4WS vehicles and the circumstances in which these systems are beneficial. Hu and Han [6] presented a comprehensive nonlinear dynamic model for 4WS vehicles. Since the previous models proposed before can be considered to be the special cases of this model, it can be used to analyze various maneuvers of 4WS vehicles. Itoh and Oida [7] studied the dynamic analysis of a 4WD-4WS vehicle and investigated speed effects on the angular velocity of the chassis rotation, as well as the sideslip angle, by the aid of related dynamics of the system. Yin et al. [8] investigated a robust controller for 4WS vehicles moving on the roads with different friction coefficients. However, no proper controlling strategy is proposed in the mentioned studies.

A four-wheeled vehicle active steering vehicle is considered as the research case in Ref. [6]. The model has eight degrees of freedom (DOFs) with nonlinear dynamic equations. Simulink is employed here to extract the simulation results. Wang and Qi [9] developed a path planning algorithm based on 4WS vehicle kinematics in which the drive angle and its velocity are two main variables. Marino et al. [10] studied the front and rear wheels' steer angles, applying the PI control method. They also noticed other parameters, such as the stable lateral velocity and the effect of lateral velocity on parking improvement. Taheri [11] controlled the rotation rate of the chassis, as well as the side speed, applying Lyapunov stability theory to the 4WS vehicle's model and considering adaptation rules. Zhu et al. [12] addressed the stability coordination control for new energy electric vehicles. They proposed a coordinated control strategy for four-wheel-distributed driving and four-wheel-distributed steering (4WD-4WS) vehicles. The strategy includes vehicle stability judgment, design of steering and yaw moment controllers, and a coordination rule based on stability indices. Experimental results demonstrate improved control effects in terms of yaw rate and side slip angle compared to a single stability control system. Shiotsuka et al. [13] proposed an adaptive controller using Neural Network in order to estimate the nonlinear friction value between tires and the road surface, leading to the control of the drive motion in 4WS vehicles. Hang and Chen [14] introduced a novel 4-wheel-steering electric autonomous vehicle. They developed a path tracking control algorithm using a single-track model and a linear parameter-varying system. The designed controller demonstrates robust and accurate path tracking performance in both simulations and real road experiments. Moreover, He and Ji [15] proposed the GCC method (global chassis control) to improve the nonlinear control of the automobile system employing its dynamic model. Akita and Satoh [16] controlled the rear wheels' steer angle using H_∞ which is a robust control method. This is considered with the instability of the trailer carrier car. However, controlling the rear-wheel steering angle is not enough on its own, while the actual nonlinear model is also approximated by a linear one. Cho et al. [17] introduced trajectory tracking controllers for a 4WS vehicle, using vehicle position and orientation as inputs. The controllers calculate wheel steering angles and velocities based on the vehicle's linear and angular velocities, enabling zero-sideslip steering maneuvers. In Ref. [18], a vehicle is controlled robustly while there is an independent control input for each wheel. The robust control strategy applied in this study is based on low-level and high-level methods, and again here, the optimization of control input is ignored considering the fact that one of the main advantages of 4WS vehicles is their high capacity of optimization since the input number is greater than its DOFs. Lee [19] have improved the stability of a vehicle, which is traveling along a straight line and is exposed to crosswise winds. This improvement is made by the aid of steering and stability booster systems. Here, the low-level controllers are combined using a parametric optimization method.

However, the optimization of the wheels' input force is

not considered in this study. Korayem et al. [20] proposed an online computing algorithm for three-wheel moving robots, which provides the optimal and smooth path for actuators in a point-to-point movement and in the presence of obstacles. Wang and Nagai [21] provided a proper operation for the absolute nonlinear behavior of the vehicle's wheels by proposing an integrated control system. He also calculated the optimum steering angles of the front and rear wheels. Lv and Liu [22] studied the robust optimal control presenting the dynamic model of 4WS vehicles and considering the rear and front wheels' steering angle to be the related control inputs. However, the required kinetic torques implemented on each wheel is not calculated. Wu et al. [23] presented a novel rear-steering based decentralized control (RDC) algorithm for four-wheel-steering (4WS) vehicles. The RDC algorithm improves lateral and directional performance, reduces sideslip magnitude, and increases yaw response bandwidth. It demonstrates robustness to uncertainties, lateral disturbances, and road friction variations, outperforming previous control methods in simulations.

Higuchi and Saitoh [24] proposed an optimal control strategy for the 4WS vehicle based on linear-quadratic regulator (LQR). Also, a similar optimal control strategy was applied by Siahkalroudi and Naraghi [25], in which zero side slip (ZSS) and zero yaw rate (ZYR) compensators are considered for the relevant linearized model. Pole placement is used by Mostavi et al. [26] to guarantee the stability of the linearized system while it is controlled by the LQR method. Furthermore, Matsumoto and Tomizuka [27] implemented the LQR method considering an extra control input in addition to the front steering angle in order to determine the status of two state variables, including the transverse and rotary motions of the vehicle. In the mentioned closed-loop optimal control studies, the Jacobian matrix is not extracted for kinematic control of the system, while this matrix is a useful tool toward controlling the mobile systems in their joint space. In other words, the desired values of the vehicle states are always in the workspace of the system while the actual states which need to be estimated using the sensors, are usually in joint space of the system, Thus, in order to calculate the related error and its consequent input, a transfer matrix, i.e., Jacobian, is required. Tourajizadeh et al. [28] calculated the optimum control inputs of a 4WS vehicle using the LQR method in order to minimize the error and energy consumption. By this means, they also derived the corresponding optimum path. The extracted controlling strategy is based on the Jacobian matrix. Lu et al. [29] introduced an adaptive LQR path tracking controller for four-wheel steering electric vehicles. It optimizes the Q and R matrices using a genetic algorithm and creates an offline table. The controller improves tracking accuracy, ensures vehicle stability, and achieves effective performance at different speeds.

An important point in the mentioned pieces of research is that LQR is a closed-loop optimization method based on linearization, which is not generally proper for a nonlinear dynamic system such as a 4WS vehicle. In other words, the linearization of such a nonlinear system in order to apply the mentioned method leads to the ignorance of some significant dynamics of the system. In another study by Korayem et al. [30], the optimum path and the control input of a mobile robot have been extracted according to Pontryagin's Minimum Principle and the calculus of variations, which is an open-loop optimization method. The open-loop optimal control method, as well as the path planning, is not yet implemented for the 4WS vehicle system. However, this study has been done before for 2WS vehicles and mobile robots. Sun et al. [31] investigate lap time improvement for a 4-wheel steer and 4-wheel independent drive vehicle in motorsports. Optimization techniques and advanced controllers helped achieve a 3% reduction in lap time by improving handling and racing lines. Authors in this study designed a path following controller, which contains a PID speed controller, as well as a model predictive control (MPC) yaw rate controller. A kinematic model is derived without the restriction relating to small-angle assumption by Tan et al. [32]. Also, an objective function and constraints of system control quantity optimization are designed based on the tracking error model. In general, this work improves path tracking for 4WS vehicles by proposing a flexible control method without restrictions on wheel steering angles. Using an optimization approach, they achieved higher tracking accuracy with reduced lateral and yaw errors compared to other algorithms. In this paper, the optimization process is considered without linearizing the system, and the nonlinear model is used to calculate the optimized control inputs.

As explained, the open-loop optimal control of a 4WS vehicle is not considered in the literature so far. Also, most of the optimization algorithms have been implemented on the linearized model of the vehicles, which is a significant simplification. Thus, in this paper, open-loop optimization and path planning of a 4WS vehicle are performed using the nonlinear model of the system. To meet this goal, the kinematic model of a 4WS vehicle is first presented based on its bicycle model. Afterward, speed constraints are derived, and dynamic equations are obtained using the Euler-Lagrange formulation. According to the extracted dynamic model, the formulation related to the proposed optimization method is extracted according to the calculus of variations. As a result, the optimum values of the control inputs, as well as the optimum path, are derived. Since the optimal path is usually useful for low-speed applications of a vehicle's movements such as parallel parking, the tire model is ignored here and it is supposed that the frictional force of the tire can provide the required propulsive force of the vehicle

and thus no slippage occurs. The verification process, in this paper, is conducted by modeling and simulating the dynamics of the 4WS vehicle system in MATLAB and applying the calculated optimal control. The correctness of modeling is investigated by comparing MATLAB results with those of CarSim. The efficiency of the proposed optimization method is also investigated by the aid of some analytic and comparative simulation scenarios. The advantages of 4WS vehicles for optimal control compared to conventional 2WS vehicles and also the superiority of using the calculus of variations method are proven. Finally, it can be concluded that employing the proposed optimal control for a vehicle in its 4WS type leads to better results, which can move the vehicle within any desired boundaries with the best accuracy and the least amount of energy consumption.

2. Modeling

2.1 Kinematics

Generally, two modeling approaches can be employed for a vehicle system. In the first one, both rear wheels are considered as a single wheel, and similarly, both front wheels are assumed to be a single one. The mentioned procedure is known as the bicycle model. The second approach is to consider each wheel of the vehicle as a separate unit to be controlled. The model is not simplified in this process, and therefore, all of the four wheels' angles are involved in dynamic equations as the joint space variables of the system. As a result, the nonlinearity, as well as the complexity of this model, is more severe than that of the bicycle model. The control analysis of this paper is performed according to the first procedure since its simplicity decreases the required online calculations and helps the system to be controlled in a real-time way. The most important effect of the studied model on extracted results is the effect of friction on vehicle tires. If the coefficient of friction between the ground and the tire would be different for different wheels of the vehicle, the complete model of the vehicle with 4 wheels can predict the slippage of different wheels of the vehicle at different parts of the system. However, in this study, it is assumed that the coefficient of friction is high enough to provide pure rolling. Thus, the locomotion process of the 4WSV can be equalized by the bicycle model with no significant simplification. The schematic view of the bicycle model is indicated in Fig. 1. The rear and front wheels' angles can be steered independently, and there are some simplifying assumptions through the process of kinematic modeling. The kinematic model is investigated considering the planar motion of the vehicle's rigid chassis in the X-Y plane and also the non-slipping condition of the wheels. According to Fig. 1, point C is the mass center of the vehicle. Point B, which is the location of the rear wheel (the

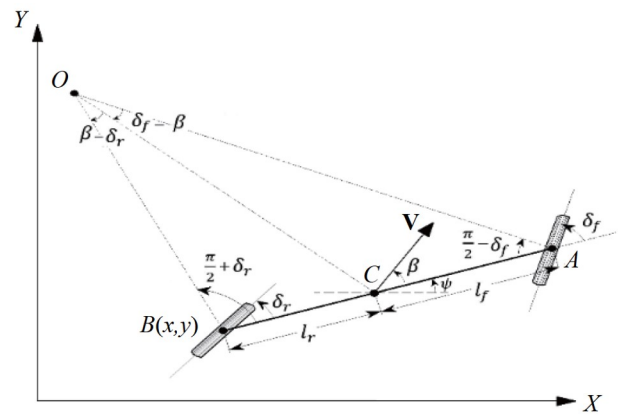

Figure 1 Bicycle model of the vehicle for kinematic modeling.

drive wheel of the vehicle), is used to show the global coordinates of the system. Vector **V** is the translational velocity of the car's center of mass. ψ is the vertex angle (yaw), which is the angle between the car's longitudinal axis and X-axis. β is the slide slip angle, which is the angle between the translational velocity of the center of mass and the vehicle's longitudinal axle. δ_f and δ_r are the front and rear wheel's steering angles, respectively. l_f is the distance between the vehicle's center of mass and the front wheel's axle. l_r is the length between the center of mass and the rear wheel's axle, and l is the length between the front and rear wheel's axles. Also, r is the wheel's radius and θ_r is the rotational angle of the rear wheel. According to Fig. 1, kinematic equations are as follows:

$$\begin{aligned}
&\dot{X} = r\dot{\theta}_r\cos(\psi+\delta_r),\\
&\dot{Y} = r\dot{\theta}_r\sin(\psi+\delta_r),\\
&\dot{\psi} = \frac{r\dot{\theta}_r\cos(\delta_r)(\tan\delta_f-\tan\delta_r)}{l},\\
&\beta = \tan^{-1}\left(\frac{l_r\tan\delta_f+l_f\tan\delta_r}{l}\right),\\
&V = r\dot{\theta}_r\cos(\delta_r)\sqrt{1+\tan\beta}\,.
\end{aligned} \tag{1}$$

It can be seen that using these equations, the workspace parameters such as X, Y, and ψ can be calculated provided that the joint space variables such as the wheels' angular velocity and the steer angles would be available. It should be noticed that the parameter ψ can be provided by the integration of $\dot{\psi}$ using an initial guess in order to achieve $\dot{X}$ and $\dot{Y}$ values. Also, in the inverse kinematic model, the joint space variables can be calculated by the workspace parameters, which is the first step to verifying extracted equations.

2.2 Kinetics

The bicycle model is again employed here to derive the kinetic model of a 4WS vehicle system. This model is shown

in Fig. 2 by its related dynamic parameters. The 4WS vehicle is an under-actuated non-holonomic system that has six DOFs and three control inputs. Also, two velocity constraints related to the rear and front wheels should be considered.

The 4WS vehicle is an under-actuated non-holonomic system that has six DOFs and three control inputs. Also, two velocity constraints related to the rear and front wheels should be considered. A 4WSV considering its kinematic constraints has six independent DOFs and its minimal state space has eight states [33].

If one supposes 7 DOFs for this system, then it is necessary to engage the related constraint which stablishes a relation between the other 6 DOFs. Another possible case for 7 independent DOFs is related to the vehicle with two drive axle systems in which both the front and rear axles can have slippage. Here it is supposed that the system has one drive axle and thus the 6 mentioned DOFs of $q = \begin{pmatrix} x & y & \psi & \theta_r & \delta_f & \delta_r \end{pmatrix}^{\mathrm{T}}$ are just independent. Equations (2) and (3) describe the velocity constraints related to the front and rear wheels of the vehicle. These constraints should be employed to realize the minimal state space of the system and extract the dynamics of the system using Lagrange multipliers. Considering these constraints and the fact that the studied vehicle of this article has one drive axle, $q = \begin{pmatrix} x & y & \psi & \theta_r & \delta_f & \delta_r \end{pmatrix}^{\mathrm{T}}$ is the configuration vector of the system where, (x, y) denotes the middle point of the rear wheels' axle. ψ is the orientation of the vehicle body. θ_r denotes the rotation angle of the rear wheel, δ_f is the front wheel's orientation, and δ_r is the rear wheel's orientation. Two mentioned velocity constraints are also as follows:

$$\begin{aligned} &\sin(\psi+\delta_f)\frac{\mathrm{d}}{\mathrm{d}t}(x+l\cos\psi)-\cos(\psi+\delta_f)\frac{\mathrm{d}}{\mathrm{d}t}(y+l\sin\psi)=0,\\ &\sin(\psi+\delta_r)\dot{x}-\cos(\psi+\delta_r)\dot{y}=0, \end{aligned} \tag{2}$$

or equivalently:

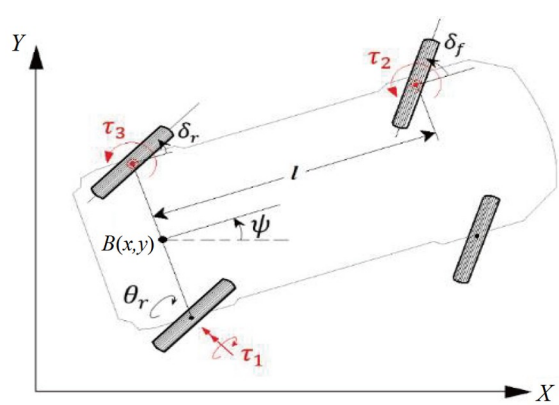

Figure 2 Bicycle model in kinetic modeling of the vehicle.

$$\begin{aligned} &\sin(\psi+\delta_f)\dot{x}-\cos(\psi+\delta_f)\dot{y}-l\dot{\psi}\cos\delta_f=0,\\ &\sin(\psi+\delta_r)\dot{x}-\cos(\psi+\delta_r)\dot{y}=0. \end{aligned} \tag{3}$$

Two recent equations can be rewritten as $\mathbf{W}(\mathbf{q})\dot{\mathbf{q}}=0$, where $\mathbf{W}=(\mathbf{W}_1,\mathbf{W}_2)$. Two matrices $\mathbf{W}_1$ and $\mathbf{W}_2$ are as Eqs. (4) and (5), which show that they are separately related to $\mathbf{q}_x=(x,y)$ and $\mathbf{q}_r=(\psi,\theta_r,\delta_f,\delta_r)$, respectively.

$$\mathbf{W}_1(\psi,\delta_f,\delta_r)=\begin{bmatrix} \sin(\psi+\delta_f) & -\cos(\psi+\delta_f)\\ \sin(\psi+\delta_r) & -\cos(\psi+\delta_r)\end{bmatrix}, \tag{4}$$

$$\mathbf{W}_2(\psi,\delta_f,\delta_r)=\begin{bmatrix} -l\cos\delta_f & 0 & 0 & 0\\ 0 & 0 & 0 & 0\end{bmatrix}. \tag{5}$$

Notice that $\mathbf{W}_1(\psi,\delta_f,\delta_r)$ is not invertible at $\delta_f=\delta_r$, we have

$$\mathbf{W}_{12}(\psi,\delta_f,\delta_r)=-\mathbf{W}_1^{-1}\mathbf{W}_2=\begin{bmatrix} \rho_1 & 0 & 0 & 0\\ \rho_2 & 0 & 0 & 0\end{bmatrix}, \tag{6}$$

where ρ_1 and ρ_2 are obtained as

$$\begin{aligned} \rho_1&=\frac{l\cos(\psi+\delta_r)\cos\delta_f}{\sin(\delta_f-\delta_r)},\\ \rho_2&=\frac{l\sin(\psi+\delta_r)\cos\delta_f}{\sin(\delta_f-\delta_r)}. \end{aligned} \tag{7}$$

Therefore, from $\mathbf{W}_{12}(\mathbf{q})$, $\mathbf{D}(\mathbf{q})$ can be obtained as follows:

$$\mathbf{D}(\mathbf{q})=\begin{bmatrix} \rho_1 & 0 & 0 & 0\\ \rho_2 & 0 & 0 & 0\\ 1 & 0 & 0 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 1\end{bmatrix}. \tag{8}$$

Now, $\mathbf{D}(\mathbf{q})$ can be used to calculate the reduced inertia matrix $\mathbf{M}_r(\mathbf{q}_r)$. The Lagrangian functional of a 4WS vehicle system is as follows:

$$\begin{aligned} \mathcal{L}=&\frac{1}{2}m\left[\left(\dot{x}-\frac{l\dot{\psi}}{2}\sin\psi\right)^2+\left(\dot{y}+\frac{l\dot{\psi}}{2}\cos\psi\right)^2\right]\\ &+\frac{1}{2}\left[2J_h\left(1+\frac{\cos^2\delta_r}{\cos^2\delta_f}\right)\dot{\theta}_r^2\right]+\frac{1}{2}(J_b+2J_v)\dot{\psi}^2\\ &+\frac{1}{2}J_v\left(\dot{\psi}+\dot{\delta}_f\right)^2+\frac{1}{2}J_v\left(\dot{\psi}+\dot{\delta}_r\right)^2, \end{aligned} \tag{9}$$

where m is the mass of the car, J_b is the body's moment of inertia, J_h is the moment of inertia related to each wheel along the horizontal axes, and J_v is their related moment of inertia along the vertical axes. The differential-algebraic equations of the motion for the system are as follows:

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial \mathcal{L}}{\partial \dot{\mathbf{q}}} - \frac{\partial \mathcal{L}}{\partial \mathbf{q}} = \mathbf{W}^{\mathrm{T}}(\mathbf{q})\begin{bmatrix}\lambda_1\\ \lambda_2\end{bmatrix} + \begin{bmatrix}\mathbf{0}_{3\times 3}\\ \mathbf{I}_{3\times 3}\end{bmatrix}\begin{bmatrix}\tau_1\\ \tau_2\\ \tau_3\end{bmatrix}, \tag{10}$$

where $\lambda_1, \lambda_2 \in \mathbb{R}$ are Lagrange multipliers, and $\tau_1, \tau_2, \tau_3 \in \mathbb{R}$ are torques applied to rear-drive wheels and steering wheels, respectively. The studied model of the 4WS vehicle system is under-actuated with non-holonomic constraints, and Eq. (10) can be reduced to a system of 8 first-order differential equations. Two of the mentioned system of equations are as below:

$$\dot{\mathbf{q}}_x = \mathbf{W}_{12}(\psi, \delta_f, \delta_r)\dot{\mathbf{q}}_r. \tag{11}$$

Also, the other 6 equations, which are shown as Eq. (12), have been extracted by eliminating Lagrange coefficients. Finally, the dynamic Eq. (10) can be rewritten to the cascade form of the constraint equation and a reduced-order Lagrangian system with configuration vector $\mathbf{q}_r = (\psi, \theta_r, \delta_f, \delta_r)$ [34].

$$\mathbf{M}_r\ddot{\mathbf{q}}_r + \mathbf{C}_r\dot{\mathbf{q}}_r + \mathbf{G}_r = \mathbf{F}_r\boldsymbol{\tau} \tag{12}$$

were

$$\begin{aligned}
\mathbf{M}_r &= \mathbf{D}^{\mathrm{T}}\mathbf{M}\mathbf{D},\\
\mathbf{C}_r &= \mathbf{D}^{\mathrm{T}}\mathbf{C}\mathbf{D} + \mathbf{D}^{\mathrm{T}}\mathbf{M}\dot{\mathbf{D}},\\
\mathbf{G}_r &= \mathbf{D}^{\mathrm{T}}\mathbf{G},\\
\mathbf{F}_r &= \mathbf{D}^{\mathrm{T}}\mathbf{F}.
\end{aligned} \tag{13}$$

Thus, forward dynamic equations can be written as follows:

$$\ddot{\mathbf{q}}_r = \mathbf{F}_r\boldsymbol{\tau} - \mathbf{M}_r^{-1}(\mathbf{C}_r\dot{\mathbf{q}}_r + \mathbf{G}_r). \tag{14}$$

3. Optimal control of the 4WS vehicle system, using the calculus of variations

Considering the fact that the system of a 4WS vehicle consists of three rigid bodies, including two front wheels, two rear wheels, and the main chassis, the state space of the system can be written as $\mathbf{q}_t = (x, \dot{x}, y, \dot{y}, \psi, \dot{\psi}, \theta_r, \delta_f, \dot{\delta}_f, \delta_r, \dot{\delta}_r)$, which includes 11 state variables for the system. According to the velocity constraints mentioned before, the number of states will decrease to 8. This outstanding result is proven in dynamic modeling after the multiplication of the null space matrix and the elimination of Lagrange multipliers. Consequently, the state-space vector can be written as follows, using kinematic equations and non-holonomic constraints.

$$\mathbf{q} = \begin{bmatrix} x & y & \psi & \delta_f & \delta_r & \dot{\theta}_r & \dot{\delta}_f & \dot{\delta}_r \end{bmatrix}^{\mathrm{T}}. \tag{15}$$

The above state-space vector is proven by theorem 6.1.1 [34].

Considering Eq. (14), u_1, u_2, u_3 can be defined as follows:

$$\begin{aligned}
u_1 &= \ddot{\theta}_r,\\
u_2 &= \ddot{\delta}_f,\\
u_3 &= \ddot{\delta}_r,
\end{aligned} \tag{16}$$

since $\ddot{\mathbf{q}}_r = \begin{bmatrix}\ddot{\theta}_r\\ \ddot{\delta}_f\\ \ddot{\delta}_r\end{bmatrix}$.

Thus, state-space equations can be written as follows:

$$\begin{aligned}
&\dot{x}_1 = rx_6\cos(x_3 + x_5),\ \dot{x}_2 = rx_6\sin(x_3 + x_5),\\
&\dot{x}_3 = \frac{r}{l}\frac{x_6\sin(x_4 - x_5)}{\cos x_4},\ \dot{x}_4 = x_7, \dot{x}_5 = x_8,\\
&\dot{x}_6 = u_1,\ \dot{x}_7 = u_2,\ \dot{x}_8 = u_3,
\end{aligned} \tag{17}$$

where $\mathbf{q}$ and $\mathbf{u}$ can be defined as

$$\mathbf{q} = [\,x_1 \quad x_2 \quad x_3 \quad x_4 \quad x_5 \quad x_6 \quad x_7 \quad x_8\,]^{\mathrm{T}}, \tag{18}$$

$$\mathbf{u} = [\,\tau_1 \quad \tau_2 \quad \tau_3\,]^{\mathrm{T}}. \tag{19}$$

The general form of the system's state space is

$$\dot{\mathbf{q}} = \mathbf{F}(\mathbf{q}, \mathbf{u}). \tag{20}$$

And the cost function is considered as follows:

$$J = \int\left(\mathbf{e}^{\mathrm{T}}\mathbf{Q}\mathbf{e} + \mathbf{u}^{\mathrm{T}}\mathbf{R}\mathbf{u}\right)\mathrm{d}t. \tag{21}$$

Moreover, the following constraints should be considered since the steering wheels have a specific rotational range due to their mechanical structure limitations:

$$-\frac{\pi}{4} \leqslant \delta_f \leqslant \frac{\pi}{4},\ -\frac{\pi}{4} \leqslant \delta_r \leqslant \frac{\pi}{4}. \tag{22}$$

According to the limitations defined in Eq. (22), the inequality constraints can be extracted as

$$\begin{aligned}
&\delta_f + \frac{\pi}{4} \geqslant 0,\ \frac{\pi}{4} - \delta_f \geqslant 0,\\
&\delta_r + \frac{\pi}{4} \geqslant 0,\ \frac{\pi}{4} - \delta_r \geqslant 0.
\end{aligned} \tag{23}$$

Therefore, the Hamiltonian function can be defined as follows:

$$\begin{aligned}
\mathbf{H} = \mathbf{V} + \boldsymbol{\lambda}_i\dot{q} + \boldsymbol{\lambda}_{i+1}\Bigg[&\left(\delta_f + \frac{\pi}{4}\right)^2\widetilde{H}\left(\delta_f + \frac{\pi}{4}\right)\\
&+\left(\frac{\pi}{4} - \delta_f\right)^2\widetilde{H}\left(\frac{\pi}{4} - \delta_f\right) + \left(\delta_r + \frac{\pi}{4}\right)^2\widetilde{H}\left(\delta_r + \frac{\pi}{4}\right)\\
&+\left(\frac{\pi}{4} - \delta_r\right)^2\widetilde{H}\left(\frac{\pi}{4} - \delta_r\right)\Bigg],\ i = 1, 2, \ldots, 8,
\end{aligned} \tag{24}$$

where

$$\mathbf{V} = \left(\mathbf{e}^{\mathrm{T}}\mathbf{Q}\mathbf{e} + \mathbf{u}^{\mathrm{T}}\mathbf{R}\mathbf{u}\right). \tag{25}$$

Also, $\boldsymbol{\lambda}_i$ refers to the Lagrange multipliers of the system and $\boldsymbol{\lambda}_{i+1}$ is related to state constraints such as steering angle limitations. Also, $\widetilde{H}$ refers to the Heaviside function, which can be expanded for this system as follows:

$$\mathbf{H} = [x_1 \quad x_2 \quad x_3 \quad x_4 \quad x_5 \quad x_6 \quad x_7 \quad x_8]\mathbf{Q}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \\ x_7 \\ x_8 \end{bmatrix}$$
$$+[\tau_1 \quad \tau_2 \quad \tau_3]\mathbf{R}\begin{bmatrix} \tau_1 \\ \tau_2 \\ \tau_3 \end{bmatrix}$$
$$+\boldsymbol{\lambda}_1 r x_6 \cos(x_3 + x_5) + \boldsymbol{\lambda}_2 r x_6 \sin(x_3 + x_5)$$
$$+\boldsymbol{\lambda}_3 \frac{r}{l}\frac{x_6 \sin(x_4 - x_5)}{\cos x_4} + \boldsymbol{\lambda}_4 x_7 + \boldsymbol{\lambda}_5 x_8 + \boldsymbol{\lambda}_6 \tau_1$$
$$+\boldsymbol{\lambda}_7 \tau_2 + \boldsymbol{\lambda}_8 \tau_3 + \boldsymbol{\lambda}_9 \Bigg[\left(\delta_f + \frac{\pi}{4}\right)^2 \widetilde{H}\left(\delta_f + \frac{\pi}{4}\right)$$
$$+\left(\frac{\pi}{4} - \delta_f\right)^2 \widetilde{H}\left(\frac{\pi}{4} - \delta_f\right) + \left(\delta_r + \frac{\pi}{4}\right)^2 \widetilde{H}\left(\delta_r + \frac{\pi}{4}\right)$$
$$+\left(\frac{\pi}{4} - \delta_r\right)^2 \widetilde{H}\left(\frac{\pi}{4} - \delta_r\right)\Bigg]. \tag{26}$$

The optimum inputs can be calculated as a function of state variables by solving the following set of simultaneous differential equations:

$$\frac{\partial H}{\partial \boldsymbol{\tau}} = 0. \tag{27}$$

This partial differential equation can be expanded as follows:

$$\frac{\partial H}{\partial \tau_1} = 0 \Rightarrow \tau_1 = \frac{1}{\cos^3 \delta_f}\Big[2J_h\Big(u_1 \cos^3 \delta_f$$
$$+u_1 \cos\delta_f \cos^2 \delta_r$$
$$+2\dot{\delta}_f \dot{\theta}_r \sin\delta_f \cos^2 \delta_r$$
$$-2\dot{\delta}_r \dot{\theta}_r \cos\delta_f \cos\delta_r \sin\delta_r\Big)\Big], \tag{28}$$

$$\frac{\partial H}{\partial \tau_2} = 0 \Rightarrow \tau_2 = J_v u_2 + J_v \ddot{\psi} - \frac{2J_h \dot{\theta}_r^2 \cos^2\delta_r \sin\delta_f}{\cos^3 \delta_f}, \tag{29}$$

$$\frac{\partial H}{\partial \tau_3} = 0 \Rightarrow \tau_3 = J_v u_3 + J_v \ddot{\psi} + \frac{2J_h \dot{\theta}_r^2 \sin(2\delta_r)}{\cos^3 \delta_f}, \tag{30}$$

substituting the optimum inputs in the following equations, the optimum states of the system are extracted, while the cost function of Eq. (21) is also optimized.

$$\begin{cases} \dfrac{\partial H}{\partial \lambda_i} = \dot{\mathbf{q}}_i, & \text{state equation,} \\ \dfrac{\partial H}{\partial \lambda_{i+1}} = \dot{\mathbf{q}}_{i+1}, & \\ -\dfrac{\partial H}{\partial \mathbf{q}_i} = \dot{\lambda}_i, & \text{co-state equation.} \\ -\dfrac{\partial H}{\partial \mathbf{q}_{i+1}} = \dot{\lambda}_{i+1}. & \end{cases} \tag{31}$$

The above set of simultaneous differential equations should be solved using the following boundary conditions:

$$x(t_0) = x_0,$$
$$\left(H^* + \frac{\partial S}{\partial x}\right)_{t_f} \delta t_f + \left[\left(\frac{\partial S}{\partial x}\right)_* - \lambda^*(t)\right]_{t_f} \delta x_f = 0. \tag{32}$$

Considering the fact that boundaries and their related traveling time are considered to be fixed, the terms δt_f and δx_f are equal to zero. The boundary condition can be summarized as follows:

$$\mathbf{x}(\mathbf{t}_0) = \mathbf{x}_0,$$
$$\mathbf{x}(\mathbf{t}_f) = \mathbf{x}_f. \tag{33}$$

4. Modeling

In this section, the accuracy of the modeling and the efficiency of the proposed optimal control are verified by the aid of CarSim and some simulation scenarios conducted in MATLAB.

4.1 Kinematic simulation

In order to verify the proposed model, a kinematic simulation is accomplished for a 4WS vehicle system using the bicycle model according to the given parameters of Table 1. The extracted kinematic model's accuracy is proven by comparing the results of the forward and inverse approaches. The desired drive wheel's rotation path, and the front and rear wheels' steering angles are considered to be

Table 1 Employed parameters in the modeling process of a 4WS vehicle

Parameter	Symbol	Value
Vehicle mass (kg)	M	1600
The inertia of the vehicle body (kg·m^2)	J_b	2300
The inertia of each wheel along its horizontal axis (kg·m^2)	J_h	0.9
The inertia of each wheel along its vertical axis (kg·m^2)	J_v	0.5
Distance from the center of mass to the front axle (m)	l_f	1.2
Distance from the center of mass to the rear axle (m)	l_r	1
Distance from the front axle to the rear axle (m)	ℓ	2.2
The radius of the wheels (m)	r	0.3

the inputs of the forward kinematic model. Outputs in this model, which are the workspace parameters of the vehicle, are extracted and considered to be the inputs of the inverse kinematic model. Therefore, the output of the forward kinematics is fed as the input of the inverse one, and the actual output of the joint variables is compared with the desired ones to check the accuracy of kinematic modeling. Equation (34) shows the desired joint space of the kinematic model.

$$\theta_r = 0.5t^2,\ \delta_f = \sin\pi t,\ \delta_r = 0.7\sin\left(\frac{\pi t}{3}\right). \tag{34}$$

The comparison between the desired and actual trajectories in the kinematic model is depicted in Fig. 3.

As can be seen, the results have excellent compatibility, which shows the correctness of kinematic modeling in the 4WS vehicle. The corresponding workspace variables, associated with the mentioned joint space, are as Fig. 4.

Moreover, V and β are shown in Fig. 5.

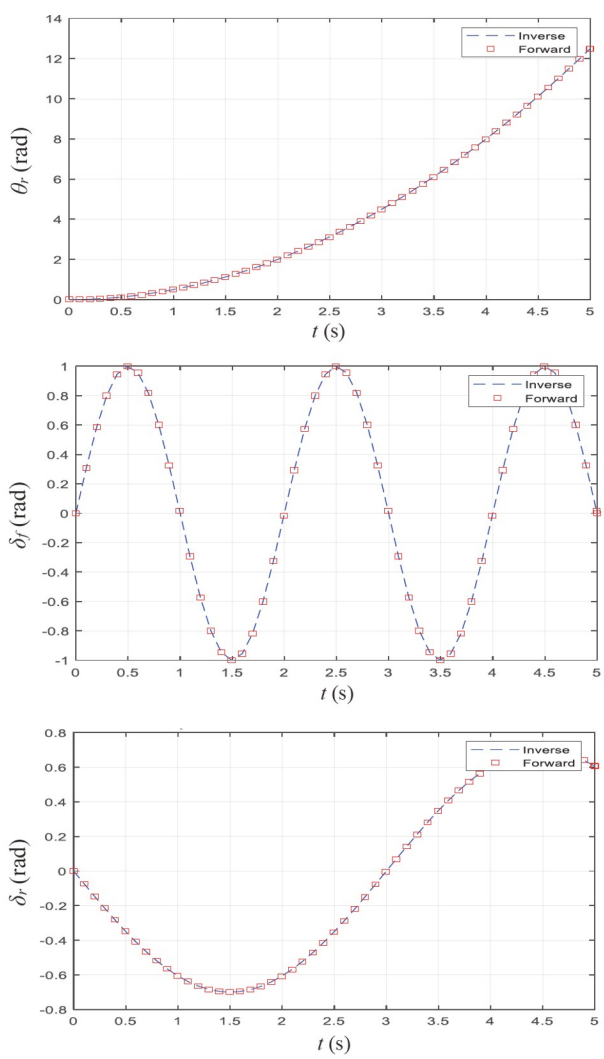

Figure 3 Comparison of the joint space paths between the forward and inverse kinematic models.

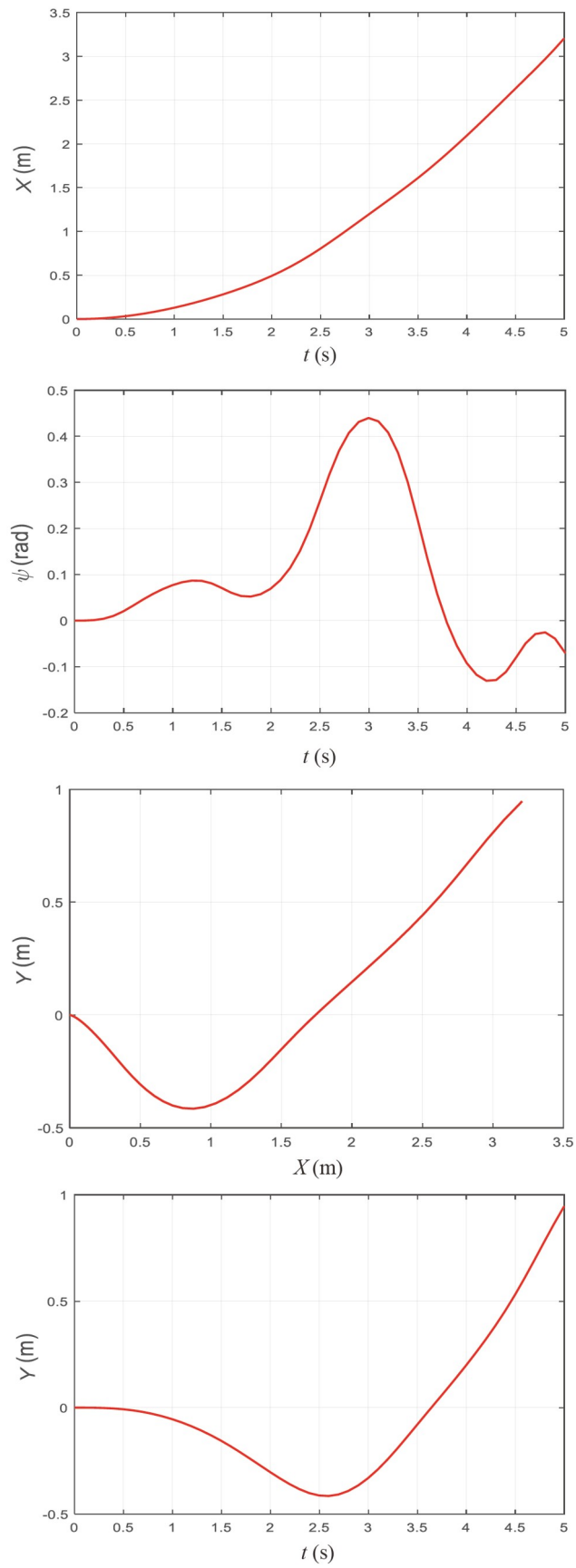

Figure 4 Corresponding workspace variables.

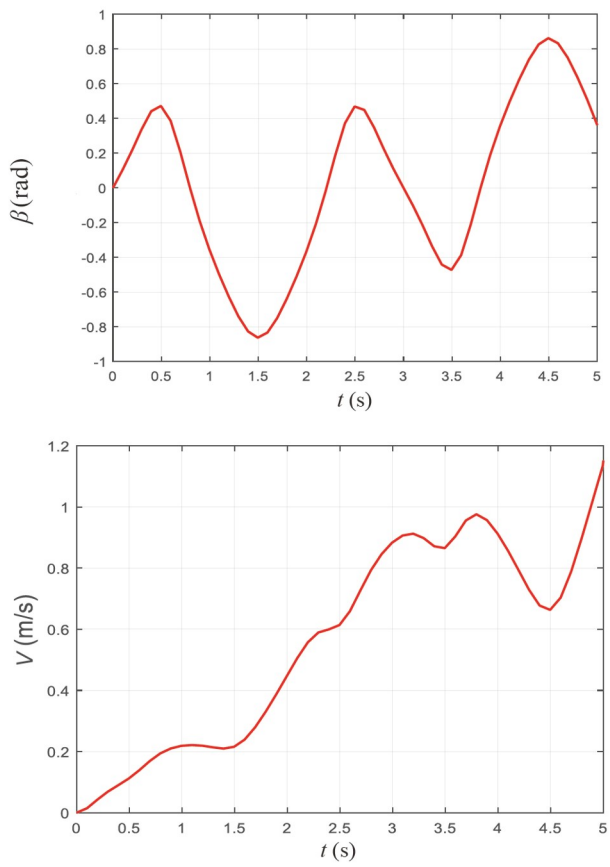

Figure 5 Velocity of the vehicle and the sideslip angle.

4.2 Kinetic simulation

A similar procedure is conducted here again in order to verify the dynamic model. In addition to the simulation parameters given in Table 1, the same desired path of Eq. (34) is considered to be the input of the dynamic model. Accordingly, the feed-forward torque amounts of the wheels are calculated using the extracted dynamic model. Afterward, the calculated inverse dynamic torques are applied to the main plant, and the actual response of the joint space variables is extracted. Therefore, the accuracy of the dynamic model can be investigated by comparing the desired and actual paths through the mentioned procedure. Figure 6 shows the results of this comparison in the kinetic model. According to Fig. 6, it can be seen that the results are appropriately matched together, and thus, it can be concluded that the kinetic model is also extracted accurately. The corresponding required torque amounts are depicted in Fig. 7 associated with the mentioned movement.

4.3 Verification and comparison with CarSim results

In order to double-check the correctness of the vehicle modeling, the simulation results are compared to those extracted by CarSim, considering the mentioned inputs of

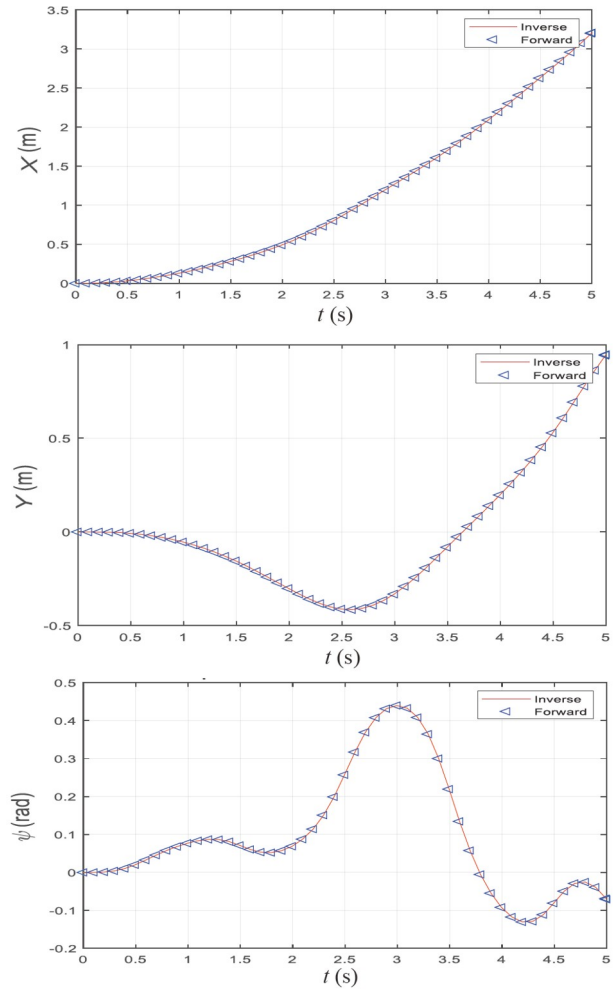

Figure 6 Comparison between the desired and actual path of the workspace variables in the kinetic model.

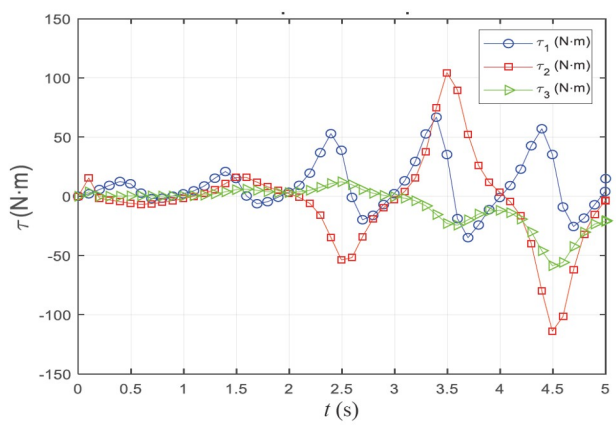

Figure 7 Required torques of the kinetic modeling.

Eq. (34), and the parameters given in Table 1. Figure 8 shows the comparison of the vehicle workspace and its related states between MATLAB and CarSim results. Here, the coefficient of friction between the tire and the road is considered 0.85 which is enough to realize the desired path of the system. The trajectory in the X-Y plane, β as the slide

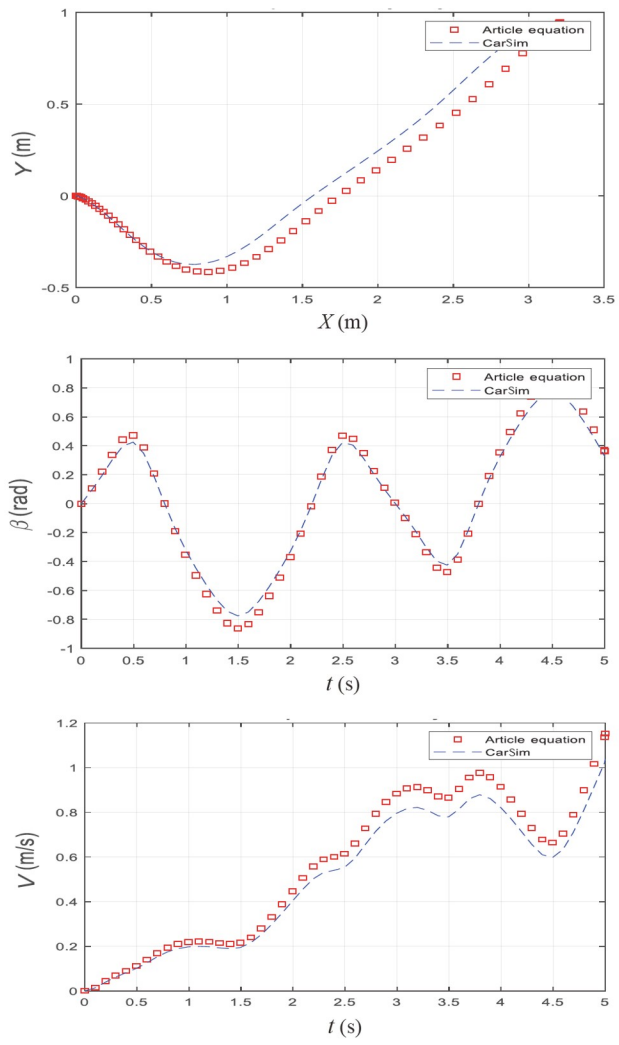

Figure 8 Comparison of the workspace and states between MATLAB and CarSim.

slip angle, and V as the vehicle speed, all in the 4WS vehicle system, are compared in Fig. 8.

It can be seen that the results are roughly the same, and the ignorable difference can be referred to as the non-modeled dynamics of the vehicle in MATLAB, which have not significant effect on results and can be ignored. These ignored dynamics include some factors such as tire modeling effects and their slippage probability, suspension system in the longitudinal and lateral dynamics of the vehicle, and aerodynamics influence of the environment, which are not modeled in MATLAB. Indeed, it is assumed that the friction between the road and the tire is sufficient to generate the required propulsion force of the vehicle toward realizing the desired optimal path. Thus, no slippage occurs and the friction force is always rotational. Comparing the results of MATLAB and CarSim shows that this assumption is roughly acceptable for the low-speed applications of the vehicle since the tire is modeled in CarSim and an ignorable slippage is occurred, which is not considerable. Also, the required torque amounts calculated by MATLAB are compared with those by CarSim, as Fig. 9 shows. Again here, the proper accordance in results shows the correctness of both the kinematics and kinetic modeling of the system in MATLAB.

4.4 Optimal control simulation by the calculus of variation

The optimal path of the modeled 4WS vehicle is planned here subject to minimizing the cost function mentioned in Eq. (21). The boundaries of the movement are predetermined, and the system is supposed to travel between these boundaries within 5 s. The following controlling gains of **Q** and **R** are considered to perform the optimal control procedure:

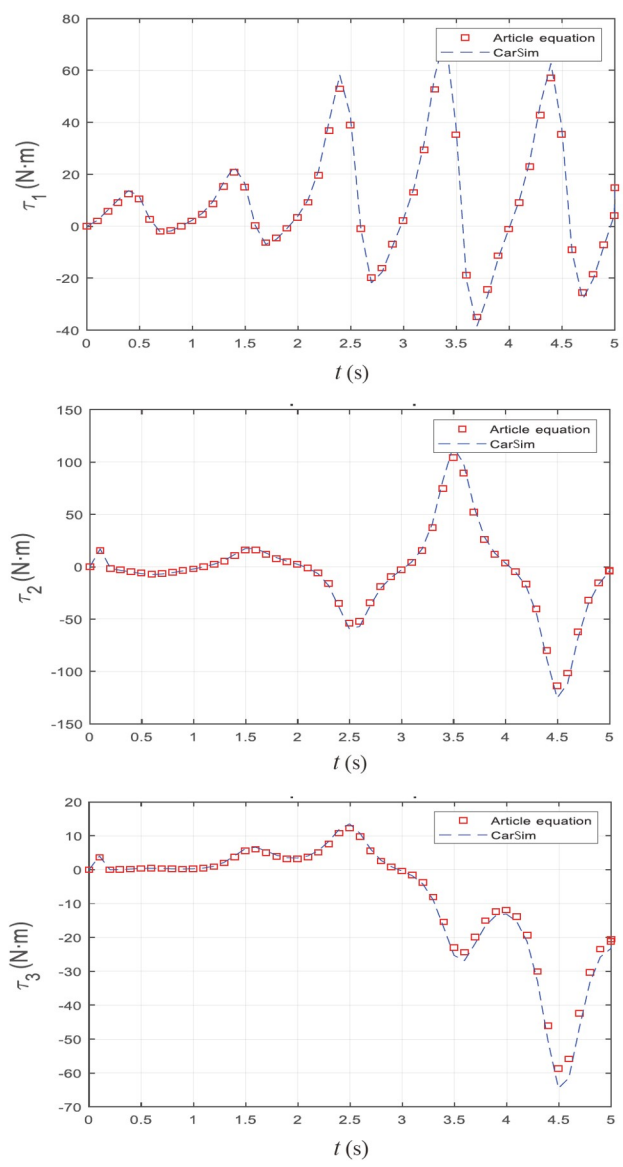

Figure 9 Comparison of the torque between MATLAB and CarSim.

$$\mathbf{Q} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{R} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}. \tag{35}$$

These controlling gains should be selected in a way that the optimum path can be generated. Therefore, they are evaluated by trial and error in a way that the maximum controlling input and the maximum allowable error bound would be saturated simultaneously. Two boundary value problems are encountered here considering the initial and final states. Solving the system of coupled differential Eq. (31) and considering the initial and final states of Eq. (36), the optimal controlling input and its related optimal path are extracted for both the 4WS and 2WS vehicle systems. The results are also compared with those related to the non-optimal approach, which has the same initial and final boundaries and a random manual path. Considering the following boundary condition for $t_f = 5$ s:

$$\begin{cases} X(t_0) = 0, \\ Y(t_0) = 0, \\ \psi(t_0) = 0, \\ q_{i+1}(t_0) = 0, \\ X(t_f) = 12.47, \\ Y(t_f) = 8.96, \\ \psi(t_f) = 0.283, \\ q_{i+1}(t_f) = 0. \end{cases} \tag{36}$$

The optimal path of 4WS and 2WS vehicles in the workspace of the system and their related optimal trajectories are shown in Fig. 10, while a new manual random path is considered here between the same boundaries for 4WS vehicle to study its differences with the mentioned optimal paths.

It can be seen that the optimal path of the 4WS vehicle is more similar to a straight line between two specified points, which is the shortest path between the boundaries. However, this optimal path is similar to a curve near the initial point as well as the endpoint, which is contributed to the zero speed of the vehicle and zero value of the yaw angle at boundary points. This result can also be seen in the optimal path of the 2WS vehicle with a bigger concussion. Therefore, it can be concluded that the optimal path of the 4WS vehicle is shorter than the two other paths.

Furthermore, Fig. 11 shows the front and rear steering angles for the optimal path of the 4WS vehicle, 2WS vehicle, and the non-optimal path. According to the figure, it can be concluded that the 4WS vehicle has the maximum amount of front and rear steering angles for each wheel. Considering the small order of these parameters, and consequently, their ignorable effect on the cost function, it will be seen that this steering input is implemented on the 4WS vehicle by the optimal controller to decrease the workspace traveling distance, and consequently, to decrease its related cost function.

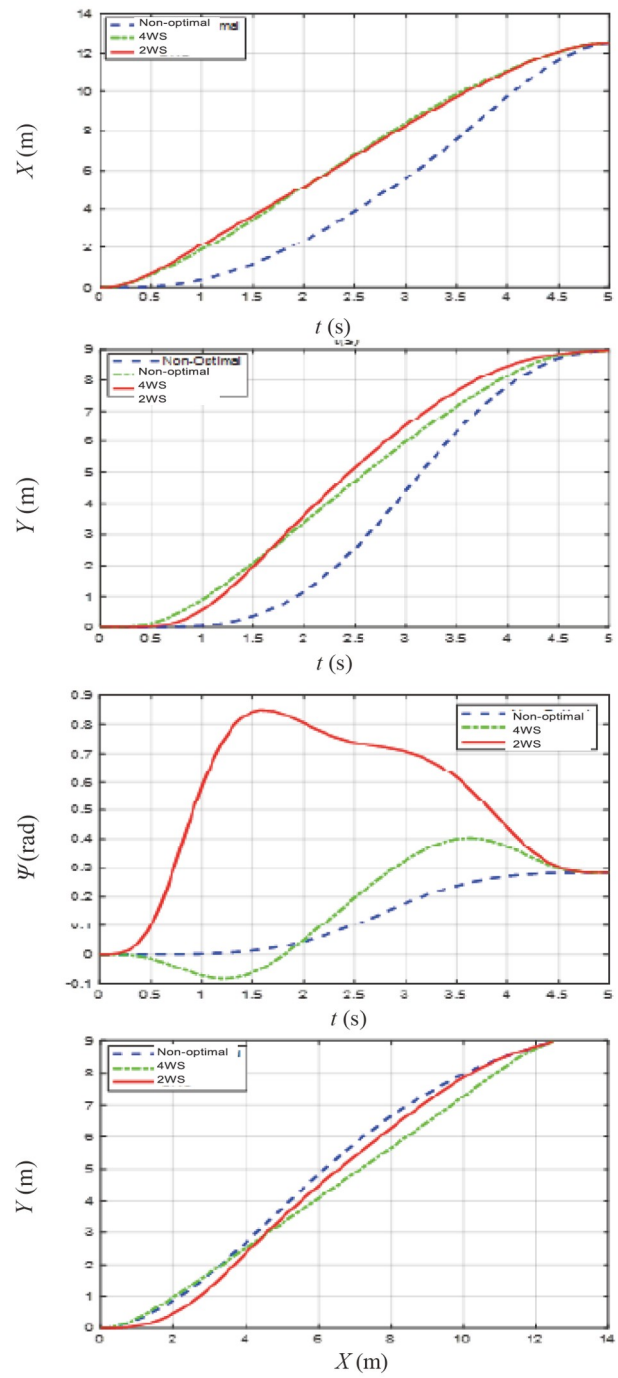

Figure 10 Comparison of the workspace variables' path and its related trajectory between the optimal 4WS, optimal 2WS, and non-optimal 4WS vehicles.

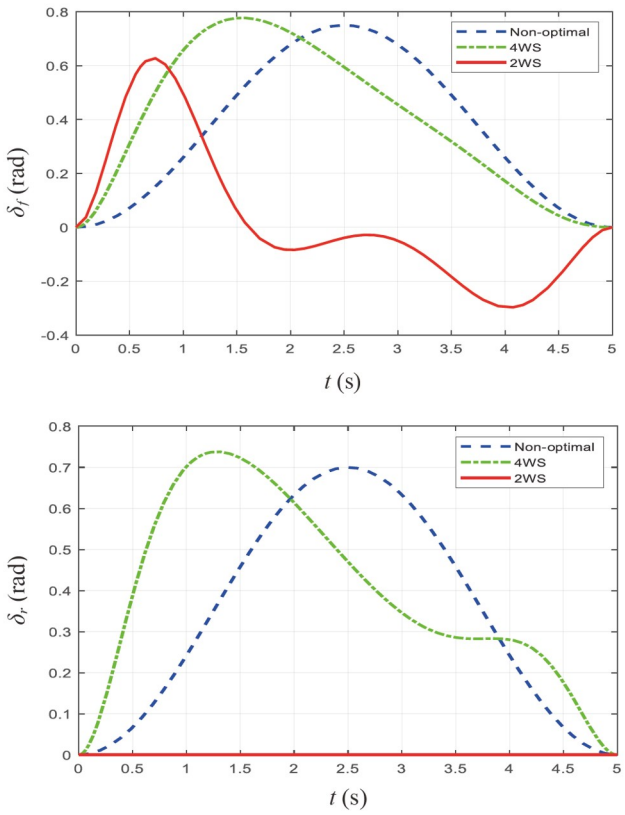

Figure 11 Comparison of the front and rear wheels' steering angle between the optimal 4WS, optimal 2WS, and non-optimal 4WS vehicles.

Figure 12 shows the vehicle's velocity in X and Y directions and also the yaw rate, respectively. It can be seen that the optimal path of the 4WS vehicle needs less acceleration and deceleration at the start and end points of the path. This verifies the fact that as expected, braking through the path should be minimum in order to maximize energy consumption.

Finally, the required torque of the drive wheel and steering wheels can be compared for the mentioned cases, as shown in Fig. 13.

According to Fig. 13, the total required amount of torque is greater for the 4WS vehicle compared to the 2WS one during the middle of traveling. However, the total amount of the overall torque during the whole movement is smaller for the 4WS system since its deceleration is much smoother than the 2WS one. This is contributed to the fact that the initial acceleration of the 2WS vehicle is more significant which should be compensated at the end of motion applying a strong brake.

Also, it should be noticed that the value of the cost function for the non-optimal case is more than those of the optimal ones, while the cost function value of the 2WS vehicle is itself bigger than that of the 4WS one. Table 2 shows the comparison of the cost function values between the applied methods of this paper.

As expected, the 4WS system decreases the cost function by about 13%. Also, the applied optimal control results in an optimal path, which significantly reduces the cost function value by about 96.4% for the 4WS vehicle.

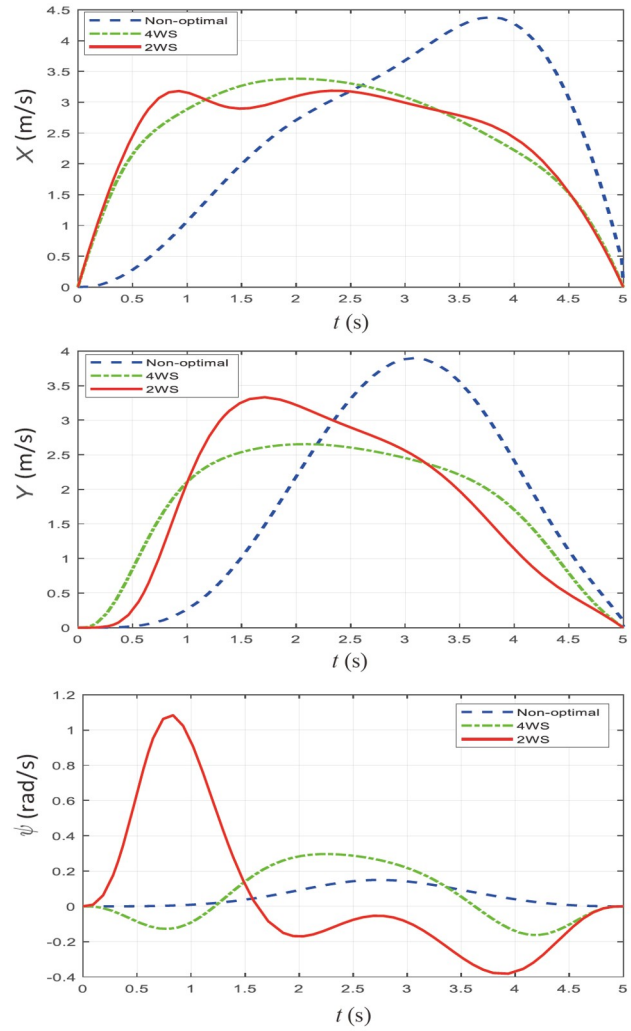

Figure 12 Comparison of the workspace variables' velocity between the optimal 4WS, optimal 2WS, and non-optimal 4WS vehicles.

5. Stability

In order to prove the stability of the system using the mentioned optimal control, first, it should be checked that the total amount of the energy in the system is constant. This is the main step to prove the stability of the system in the absence of any disturbance. Assuming the Hamiltonian function introduced by Eqs. (27) and (31), if and only if the smooth functions of $\boldsymbol{\lambda}_i(t) : [t_0, t_f] \to \mathbb{R}^n$ and $u_i(t) : [t_0, t_f] \to \mathbb{R}^n$ exist, which comply with Eqs. (27) and (31) by $x_i(t) : [t_0, t_f] \to \mathbb{R}^n$, then $x(t)$ will comply with the Euler-Lagrange equations for the Lagrangian equation. The

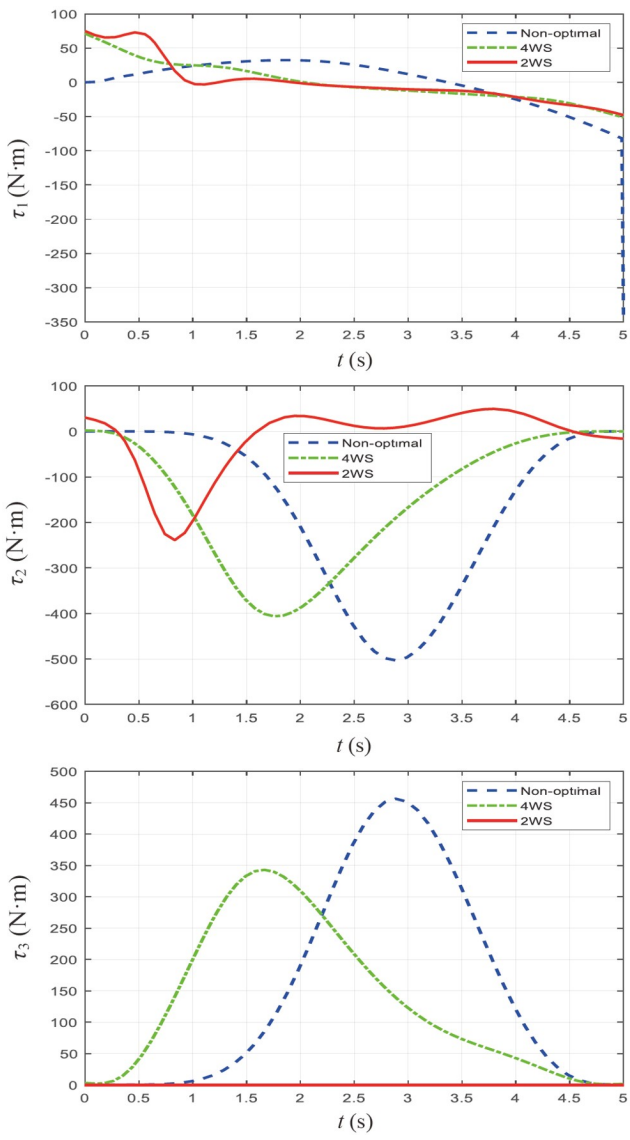

Figure 13 Comparison of the input torque amounts between the optimal 4WS, optimal 2WS, and non-optimal 4WS vehicles.

Table 2 Comparison of the cost function values between the three mentioned methods

Control method		Cost function value
Optimal	4WS	1.6585×10^3
	2WS	1.9043×10^3
Non-optimal		4.5980×10^4

interesting point is that the Hamiltonian function is the extremum of the energy function here. Moreover, it can be indicated that

$$\frac{\mathrm{d}H}{\mathrm{d}t}\left(t,\mathbf{x}(t),\dot{\mathbf{x}}(t),\frac{\partial L}{\partial \dot{\mathbf{x}}}(t,\mathbf{x}(t),\dot{\mathbf{x}}(t))\right) = \frac{\partial H}{\partial t}\left(t,\mathbf{x}(t),\dot{\mathbf{x}}(t),\frac{\partial L}{\partial \dot{\mathbf{x}}}(t,\mathbf{x}(t),\dot{\mathbf{x}}(t))\right). \tag{37}$$

The above equation is extracted using the chain rule for derivatives, as well as Eqs. (27) and (31), which is obtained by the definition of the Hamiltonian function. Thus, the Hamiltonian function is shown as follows, according to the Lagrangian [35].

$$L = -H + \sum_{i=1}^{n} \lambda_i^{\mathrm{T}} \dot{\mathbf{x}}_i. \tag{38}$$

According to the definition of Eq. (38), it can be concluded that

$$\frac{\partial H}{\partial t}\left(t,\mathbf{x}(t),\dot{\mathbf{x}}(t),\frac{\partial L}{\partial \dot{\mathbf{x}}}(t,\mathbf{x}(t),\dot{\mathbf{x}}(t))\right) = -\frac{\partial L}{\partial t}(t,\mathbf{x}(t),\dot{\mathbf{x}}(t)). \tag{39}$$

According to Eqs. (37) and (39), since the Lagrangian is defined as the difference between the potential and kinetic energies of the system $L = (U - V)$, it can be concluded that the Lagrangian is not explicitly related to time, and the left side of Eq. (39) is equal to zero.

$$\frac{\partial H}{\partial t} = 0. \tag{40}$$

Consequently, one can conclude that the Hamiltonian function, or the energy function, is constant during the path. Thus, the system is stable [36].

In order to find the optimum path between two specific points according to a determined cost function, it should be considered that the state's error, as well as the input, is decreasing over the time until the end of the route, in which the vehicle is completely stopped. Therefore, the function in Eq. (21) can be assumed as a positive definite function introduced in the Lyapunov Theory, to prove the stability of the system. With the above assumptions about the state errors and the input, it is clear that the function in Eq. (21) is decreasing over the time, which is the stability proof according to the Lyapunov theory. Finally, using the PID control method the robustness of the optimum response can be guaranteed even against the external disturbances [37].

6. Conclusion

The optimal control of a 4WS vehicle was conducted in this paper, and its related optimal path was extracted. After deriving the corresponding kinematic equations, the dynamic model of the system was extracted using the Euler-Lagrange method considering non-holonomic constraints and the elimination of Lagrange multipliers. Afterward, the Hamiltonian function was formed based on the cost function and considering dynamic constraints. Finally, the required set of simultaneous differential equations related to the calculus of variations was extracted through which the optimal control input and optimal path can be calculated. Besides, according to the obtained diagrams and presented formulations, it was proven that the calculus of variations

approach is stable in the absence of disturbance and this stability is due to the use of optimization methods based on the Euler-Lagrange equation of the Hamilton's function.

Solving the mentioned equations with predefined boundary value conditions, the optimal input, and its corresponding optimal path were extracted for transferring the vehicle from an arbitrary starting point to a final target. The correctness of the model was verified by MATLAB software and by comparing its results with those of CarSim. Furthermore, the efficiency of the proposed optimization method and its related equations were investigated by conducting some analytic and comparative simulation scenarios. The Kinematics and kinetics of the model are verified by comparing their relevant inverse and direct phases. It is seen that about 96.4% improvement can be realized in the energy consumption of a 4WS vehicle, which is controlled using the proposed optimization approach. It can be observed that the extracted optimal paths have the Gaussian shape in velocity profiles and S shape in position profiles, and the mentioned optimization is the result of these smooth generated paths. According to Fig. 13, the required torque for the drive wheel, which is calculated by the optimization method, has a better path compared to the non-optimal mode, since the optimum path is smoother than the non-optimal one. Also, the comparison between 2WS and 4WS vehicles indicates a 13% improvement for 4WS vehicles, which shows their maneuverability superiority compared to conventional 2WS vehicles. Thus, it can be concluded that the proposed optimal control can transfer a 4WS vehicle between any arbitrary boundaries with the best accuracy and the least amount of energy consumption.

Conflict of interest *On behalf of all authors, the corresponding author states that there is no conflict of interest.*

Author contributions *Conceptualization was performed by Hami Tourajizadeh and this author designed the research. Formal analysis, Investigation, and Methodology were done by Hami Tourajizadeh and Mohsen Sarvari. Supervision and Project administration were performed by Hami Tourajizadeh. Software and Validation were accomplished by Mohsen Sarvari. Hami Tourajizadeh and Mohsen Sarvari wrote the first draft of the manuscript. Visualization, Writing – original draft, and Writing – review & editing were accomplished by all of the authors. Samira Afshari helped organize the manuscript. Hami Tourajizadeh , Mohsen Sarvari, and Samira Afshari revised and edited the final version.*

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基于变分法的四轮转向机动车辆路径规划及最优控制研究

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摘要 使用变分法对四轮转向机动车辆(4WSV)进行最优控制研究, 并根据预定目标函数提取出相对最优路径. 当前, 4WS车辆因为其出色的操控性能增长迅速. 由于4WS车辆系统是过约束的, 因此车辆在两个确定边界之间可能轨迹并不唯一. 由此, 通过最小化预定成本函数提取最优路径成为一项有益的研究. 为了达到此目的, 本文首先确定系统模型, 随后应用变分法来进行车辆的最优控制. 控制输入的最优值不仅满足边界条件, 同时实现特定成本函数的最小化. 成本函数被设定为最小化误差和能量消耗, 并对相应最优路径也进行了计算. 为了验证模型的正确性, 本文将MATLAB的仿真结果与CarSim进行了比较. 通过在MATLAB中进行解析和比较性模拟场景的分析, 研究最优控制方法的效率. 通过比较相应的成本函数, 证明了本文提出的控制方法相较非最优策略的优越性. 最后证明本文提出的最优控制方法可以获得两个预定边界之间的最优路径, 并通过该使用变分法对四轮转向机动车辆(4WSV)进行最优控制研究, 并根据预定目标函数提取出相对最优路径. 当前, 4WS车辆因为其出色的操控性能增长迅速. 由于4WS车辆系统是过约束的, 因此车辆在两个确定边界之间可能轨迹并不唯一. 由此, 通过最小化预定成本函数提取最优路径成为一项有益的研究. 为了达到此目的, 本文首先确定系统模型, 随后应用变分法来进行车辆的最优控制. 控制输入的最优值不仅满足边界条件, 同时实现特定成本函数的最小化. 成本函数被设定为最小化误差和能量消耗, 并对相应最优路径也进行了计算. 为了验证模型的正确性, 本文将MATLAB的仿真结果与CarSim进行了比较. 通过在MATLAB中进行解析和比较性模拟场景的分析, 研究最优控制方法的效率. 通过比较相应的成本函数, 证明了本文提出的控制方法相较非最优策略的优越性. 最后证明本文提出的最优控制方法可以获得两个预定边界之间的最优路径, 并通过该路径可以实现最大准确性和最小能量消耗.